



Lama Alsurakhi (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

20

out of 20 points

Dry Air (with molar composition 21% O₂ and 79% N₂) is compressed from P₁=1 bar and T₁=30 °C to a final pressure P_{final}=100 bar in a two stage compressor with intermediate cooling of the first stage discharge stream down to T₃=30 deg C before entering the second stage. The two stages have the same pressure ratio and isentropic efficiency. A thermocouple measures the temperature of the second stage discharge stream to be equal to T_{final}=365 °C. The molar flow rate of the Dry Air is 1.5 kmol/s and its thermal properties are described by the **Peng-Robinson** Equation of State with all Binary Interaction Parameters (BIPs) set equal to zero. Calculate the total power requirement for this process. Express your result in **MW** and round your numerical answer **into a single decimal digit**.



Given Answer: 30.7

Correct Answer: 30

Answer range +/- 1 (29 - 31)

QUESTION 2: CALCULATED NUMERIC

20

out of 20 points

A liquid solution of n-Pentane (C₅H₁₂) [Component **1**], n-Hexane (C₆H₁₄) [Component **2**] and Cyclohexane (C₆H₁₂) [Component **3**] is initially at T=25 °C and P=2 bar. The molar composition of the mixture is [Z₁,Z₂,Z₃]=[0.40,0.15,0.45] and the volumetric flow rate 1000 gpm. The mixture is subsequently heated(isobarically) until it reaches its boiling point in a heat exchanger using saturated steam (with quality equal to one) at P=2.0 bar. Calculate the minimum steam flow rate required for the heating process. Assume that both phases (vapor and liquid) of solution follow the **Peng-Robinson** Equation of State with Binary Interaction Parameters (BIPs) all set equal to zero. Express your result in **MMSCFD** and round your numerical answer **into two decimal digits**.



Given Answer: 9.58

Correct Answer: 9.15

Answer range +/- 0.85 (8.30 - 10.00)

QUESTION 3: CALCULATED NUMERIC

20

out of 20 points

Cold liquid **LPG** (Liquefied **P**etroleum **G**as) at pressure $P_1=10$ bar and temperature $T_1=5^\circ\text{C}$, with molar composition: Propane=0.60, n-Butane=0.3 and i-Butane=0.1, is throttled adiabatically down to a pressure of $P_2=1$ bar. Calculate the Lost Work of the throttling operation, if the flow rate of **LPG** is 100 kmol/hr. Assume that the thermal properties of the **LPG** are described by the **Peng-Robinson** equation of state with Binary Interaction Parameters (BIPs) all set equal to zero. Express your answer in **kW** and round your numerical result into **a single decimal digit**.



Given Answer: ☒ 13.2

Correct Answer: ☒ 12.6

Answer range +/- 1 (11.6 - 13.6)

QUESTION 4: CALCULATED NUMERIC

20

out of 20 points

A fluid follows the following equation of state: $\left(P - \frac{a}{V^2}\right)V = RT$ where a is a constant that depends on the critical properties of the fluid. Calculate the Residual Gibbs Free Energy of this fluid at $T=30^\circ\text{C}$ and $P=10$ bar, if the compressibility factor at these conditions is $Z=0.88$. Express your result in **kJ/kmol** and round your numerical answer **into whole number**.



Given Answer: ☒ -283

Correct Answer: ☒ -283

Answer range +/- 1 (-284 - -282)

QUESTION 5: CALCULATED NUMERIC

10

out of 10 points

Calculate the fugacity of dry ice (solid CO_2) at the triple point ($T=216.55$ K, $P=5.12673$ bar). Assume that the thermal properties of the vapor phase of carbon dioxide(CO_2) are described by the **Peng-Robinson** equation of state. Express your result in **bar** and round your numerical answer **into a single decimal digit**.



Given Answer: ☒ 4.8

Correct Answer: ☒ 4.8

Answer range +/- 0.1 (4.7 - 4.9)

QUESTION 6: CALCULATED NUMERIC

10

out of 10 points

Calculate the **Internal Energy** of the Saturated Vapor of n-Octane in **kJ/mol** at $T=180^\circ\text{C}$. The reference state is $h = 0$ at the normal boiling point for saturated liquid. Round your numerical answer **into a single decimal digit**. Assume that both the vapor and liquid phases of n-Octane follow the **Peng-Robinson** Equation of State.



Given Answer: ☒ 44.1

Correct Answer: ☒ 44.1

Answer range +/- 0.1 (44.0 - 44.2)